Excluding the Bounded Negative Part

Lean-checked rigidity for Erdős Problem #243

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Abstract

Sylvester's sequence $2, 3, 7, 43, 1807, \ldots$ satisfies $a_n = a_{n-1}^2 - a_{n-1} + 1$ and has reciprocal sum 1. Erdős asked whether a rapidly growing integer sequence with rational reciprocal sum must satisfy that recurrence eventually. Clearing denominators gives integer states $D_{n+1} = a_n D_n$ and $C_{n+1} = a_n C_n - D_n$, and centring at the Sylvester tail gives $E_n = D_n - (a_n - 1)C_n$, with $C_{n+1} = C_n - E_n$; the question becomes whether E can avoid vanishing. When E is nonnegative a one-line descent settles it. This note is about the negative side.

We formalise in Lean that E cannot be negative in any of four increasingly general ways. It cannot be constantly $-m$, at any magnitude and any scale: every multiplier would have to share a prime with the fixed m, and once a prime divides one multiplier every later multiplier is 1 modulo it, so a fixed integer would need infinitely many distinct prime divisors. It cannot be periodic with positive drift, by the same pigeonhole after a strong induction that removes common scale. And it cannot have a bounded negative part at all: a bounded negative part makes C rise by a bounded amount, normalised vanishing makes C tend to infinity, the tail gcd stabilises so the orbit may be taken reduced, and in a reduced tail the multipliers are pairwise coprime and permanently avoided by the numerator, which a Chinese-remainder block of consecutive multiples forbids. Together with an absorption lemma making $E_n = 0$ propagate, this yields: under exact dynamics, eventual strict centring, an eventually bounded negative part, and normalised vanishing, E vanishes eventually, and the sequence satisfies the Sylvester recurrence eventually.

Problem #243 is open and this note does not close it. The last theorem is conditional, and two of its hypotheses are analytic inputs that we assume rather than derive. What the results remove is the bounded case: any counterexample must have negative excursions that are unbounded along a cofinal set of indices.

1 The problem

Sylvester's sequence $2, 3, 7, 43, 1807, \ldots$ is generated by $a_n = a_{n-1}^2 - a_{n-1} + 1$ and has reciprocal sum 1; the recurrence is exactly the statement that the tail beyond a_{n-1} equals $1/(a_{n-1} - 1)$. Erdős Problem #243 asserts that this is the only way a sequence of that growth can have a rational reciprocal sum:

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Problem 1.1 (Erdős #243)**.** Let $1 \le a_1 < a_2 < \cdots$ be a sequence of integers with

$$\lim_{n\to\infty} \frac{a_n}{a_{n-1}^2} = 1 \qquad \text{and} \qquad \sum \frac{1}{a_n} \in \mathbb{Q}.$$

Then $a_n = a_{n-1}^2 - a_{n-1} + 1$ for all sufficiently large n.

See [3, p. 64] and [4, p. 105]; numbering and status follow Bloom's catalogue [5]. The problem is open. Below we index the same recurrence as $a_{n+1} = a_n^2 - a_n + 1$, which is the shift the formal sources use; the two forms are the same statement. Write $\mathrm{syl}(a) = a^2 - a + 1$, the Sylvester successor.

Relation to prior work. Two results bound what is new here, and both are stronger than anything below. Erdős and Straus proved that if $\lim a_n/a_{n-1}^2 = 1$, the reciprocal sum is rational, and a_n does not satisfy the recurrence, then

$$\limsup_{n\to\infty} \frac{[a_1, \dots, a_n]}{a_{n+1}} \left(\frac{a_n^2}{a_{n+1}} - 1 \right) > 0,$$

where $[a_1, \dots, a_n]$ is the least common multiple [1]. Duverney proved a conditional form of the problem itself: if $\sum_{n\ge 0}(a_{n+1}/a_n^2 - 1)$ converges, then the reciprocal sum is rational if and only if the recurrence holds for all large n [2]. Neither is formalised here, and the results below do not recover either of them. The statement of Problem #243 has also been formalised before, as a conjecture with an unfilled proof, in the *formal-conjectures* collection; that is a statement of the question. No claim of priority is made for anything below.

Why a state system. The hypothesis is asymptotic and the conclusion is exact, so the argument must convert an analytic rate into an integer obstruction. The conversion used here is the classical one: clear denominators along the sequence, so that rationality makes a tail integral, and then centre that integer at the value it would take on Sylvester's sequence. What remains is a single integer error E, and the whole question is whether E can avoid vanishing. Section 4 disposes of the case $E \ge 0$ in three lines. The rest of the note is the case $E < 0$, in four widening steps: constant magnitude (Section 5), periodic magnitude (Section 6), bounded magnitude via a coprimality barrier (Sections 7 and 8), and what is left (Section 10).

Status of everything below. *Status.* The problem treated here is open, and this note does not close it. Every statement below marked as checked is a proposition that the pinned Lean kernel accepts from the sources this note links to, with no `sorry`, no added axiom, and no unchecked evaluation. That is a claim about the formal statement, not about its mathematical interest, its novelty, or the original problem. The unresolved obligations are named exactly, in their own section, and none of the finite computations, reductions, or no-go results here removes one of them.

Reading map. Sections 2–4 set up the state system and settle the nonnegative case. Sections 5–8 are the negative case and contain the new content; a reader who wants only the strongest statement should read Section 8 and then Section 10. Linked phrases open the corresponding Lean declaration at the pinned source revision 4eb4c8ecfa0a.

Statement	Status	Treatment here
Problem #243 as stated	Open	Not proved.
Defect identity	Proved here	Theorem 3.1, an identity in $\mathbb{Z}[a, a', D, C]$.
Eventual vanishing forces the recurrence	Proved here	Theorem 3.3.
Zero is absorbing	Proved here	Theorem 3.4, under strict centring.
Nonnegative E	Proved here	Theorem 4.1, by descent.
Constantly negative E	Excluded here	Theorem 5.1, at any magnitude and scale.
Periodic negative magnitude	Excluded here	Theorem 6.1, in the regime $e_n < a_n$.
Coprime-modulus barrier	Proved here	Theorem 7.2.
Bounded negative part	Excluded here; conditional	Theorem 8.1; see its hypotheses.
Strict centring, normalised vanishing	Assumed, not derived	Section 8.
Unbounded negative excursions	Open	Section 10; the surviving obstruction.
Erdős–Straus criterion; Duverney	Proved elsewhere	Cited, not formalised; both stronger than anything here.
State system equals reciprocal tails	Not formalised	Section 2.

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2 The centred integer state

For $a, D, C \in \mathbb{Z}$ define

$$\mathrm{den}(a, D) = aD, \quad \mathrm{tail}(a, D, C) = aC - D, \quad \mathrm{ctr}(a, D, C) = D - (a-1)C,$$

the denominator update, the tail update, and the centred state. Given (a_n) we write $D_{n+1} = \mathrm{den}(a_n, D_n)$, $C_{n+1} = \mathrm{tail}(a_n, D_n, C_n)$ and $E_n = \mathrm{ctr}(a_n, D_n, C_n)$.

Proposition 2.1 (update law)**.** $\mathrm{tail}(a, D, C) = C - \mathrm{ctr}(a, D, C)$; *that is,* $C_{n+1} = C_n - E_n$.

Proof. $C - (D - (a-1)C) = aC - D$. □

Formalised as the update law. The state therefore moves by exactly the error at each step, which is what makes the sign of E decisive: where E is positive the state falls, where E is negative it rises, and where E vanishes it is stationary.

The intended reading, which is not formalised. Let $T_n = \sum_{k \geq n} 1/a_k$ and suppose $T_0 \in \mathbb{Q}$. Put D_0 equal to a common denominator and $D_n = D_0 a_0 \cdots a_{n-1}$, and set $C_n = D_n T_n$. Then $C_n \in \mathbb{Z}$ for every n, and from $T_n = 1/a_n + T_{n+1}$ one gets $C_{n+1} = a_n C_n - D_n$, which is the tail update. On Sylvester's sequence $T_n = 1/(a_n - 1)$ exactly, so $D_n =$

$(a_n - 1)C_n$ and $E_n = 0$: the centred state measures deviation from the Sylvester tail identity, and it vanishes identically on the Sylvester orbit.

None of this paragraph is formalised. The Lean module contains no series, no rationality hypothesis, and no growth hypothesis; it proves identities and implications about the integer system above, and about its natural-number realisation, for which the bridging identities are the tail realisation, the denominator realisation, and the signed update law. A reader should treat the identification with reciprocal tails as motivation for the definitions and not as a checked step.

Proposition 2.2 (scale equivariance)**.** *For every $s, a, D, C \in \mathbb{Z}$,*

$$\operatorname{den}(a, sD) = s \operatorname{den}(a, D), \qquad \operatorname{tail}(a, sD, sC) = s \operatorname{tail}(a, D, C),$$

$$\operatorname{ctr}(a, sD, sC) = s \operatorname{ctr}(a, D, C).$$

Formalised as the denominator equivariance, the tail equivariance, and the centred equivariance. The system depends only on the ray through (D, C). This is used twice below, once to reduce a search and once, in Theorem 6.1, as the induction step that removes common scale.

3 The defect identity, rigidity, and absorption

Write $\operatorname{def}(a, a') = a' - \operatorname{syl}(a)$ for the deviation of the next term from the Sylvester successor, the Sylvester defect.

Theorem 3.1 (defect identity)**.** *For all $a, a', D, C \in \mathbb{Z}$,*

$$\operatorname{def}(a, a') \cdot \operatorname{tail}(a, D, C) = a^2 \operatorname{ctr}(a, D, C) - \operatorname{ctr}(a', \operatorname{den}(a, D), \operatorname{tail}(a, D, C)),$$

that is, $\Delta_n C_{n+1} = a_n^2 E_n - E_{n+1}$.

Proof. Both sides expand to $a'aC - a'D - a^3C + a^2D + a^2C - aD - aC + D$. □

Formalised as the defect identity. It is a polynomial identity in four indeterminates, with no hypothesis of any kind, and it carries the whole argument twice over: once for rigidity, and once for absorption.

Theorem 3.2 (two vanishing errors force the step)**.** *Let $a, a', D, C \in \mathbb{Z}$ with $\operatorname{tail}(a, D, C) \neq 0$. If $\operatorname{ctr}(a, D, C) = 0$ and $\operatorname{ctr}(a', \operatorname{den}(a, D), \operatorname{tail}(a, D, C)) = 0$, then $a' = \operatorname{syl}(a)$.*

Proof. By Theorem 3.1 the product $\operatorname{def}(a, a') \cdot \operatorname{tail}(a, D, C)$ is zero, and the second factor is nonzero by hypothesis. □

Formalised as the local rigidity step. The hypothesis $C_{n+1} \neq 0$ is not removable: at $C_{n+1} = 0$ the identity gives no information about a'.

Theorem 3.3 (sequence form)**.** *Let $a, D, C : \mathbb{N} \to \mathbb{Z}$ satisfy $D_{n+1} = \operatorname{den}(a_n, D_n)$ and $C_{n+1} = \operatorname{tail}(a_n, D_n, C_n)$. If $E_n = 0$ for all sufficiently large n and $C_{n+1} \neq 0$ for all sufficiently large n, then $a_{n+1} = a_n^2 - a_n + 1$ for all sufficiently large n.*

Formalised as the eventual Sylvester recurrence: take the larger of the two thresholds and apply Theorem 3.2 at each later index. This is the theorem every later section feeds: from here on the goal is exactly to force E to vanish eventually.

The defect identity has a second consequence, which is what makes a single vanishing error worth having.

Theorem 3.4 (zero is absorbing)**.** *Let $a, C, D : \mathbb{N} \to \mathbb{N}$ be an exact natural orbit, so $C_{n+1} + D_n = a_n C_n$ and $D_{n+1} = a_n D_n$, and let $E_n = \mathrm{ctr}(a_n, D_n, C_n)$. Suppose the centring is strict, $|E_n| < C_n$ for every n. If $E_n = 0$ then $E_{n+1} = 0$.*

Proof. Putting $E_n = 0$ in Theorem 3.1 gives $\Delta_n C_{n+1} = -E_{n+1}$, so C_{n+1} divides E_{n+1}. A multiple of C_{n+1} of absolute value smaller than C_{n+1} is zero, and $|E_{n+1}| < C_{n+1}$ by hypothesis. □

Formalised as the absorption of a vanishing centred state, with the companion contrapositive along a tail: if zero is absorbing beyond some index and E does not vanish eventually, then E is nowhere zero beyond that index.

Absorption changes the shape of the problem. Without it, one would have to exclude an E that dips to zero and recovers. With it, either E hits zero once and is finished, or it is never zero at all; the later sections may therefore assume that $E_n \neq 0$ for every n, and in the negative case that $E_n < 0$ throughout, without loss.

4 The nonnegative case

Theorem 4.1 (descent)**.** *Let $C, E : \mathbb{N} \to \mathbb{N}$ satisfy $C_{n+1} + E_n = C_n$ for every n. Then $E_n = 0$ for all sufficiently large n.*

Proof. The relation forces $C_{n+1} \leq C_n$, so C is a nonincreasing sequence of natural numbers and is eventually equal to its minimum; beyond that index $C_{n+1} = C_n$, and the relation gives $E_n = 0$. □

Formalised as the stabilisation of the nonnegative error, on the eventual constancy of a nonincreasing sequence.

The hypothesis is exactly one-sidedness: C and E take values in $\mathbb{N}$. If $E_n < 0$ at infinitely many indices then C rises there, nothing descends, and the argument is unavailable. That is the whole difficulty, and the rest of this note attacks it.

5 Constant negative magnitude

Suppose $E_n = -m$ for a fixed $m > 0$ and every n. By Proposition 2.1 the state then climbs by exactly m each step, so $C_n = c + nm$ with $c = C_0$, and the definition of the centred state becomes a *shape equation*

$$D_n + m = (a_n - 1)\,(c + nm). \tag{5.1}$$

Together with $D_{n+1} = a_n D_n$ this is a closed system in (a, D), and it has no solutions.

Theorem 5.1 (no constant negative magnitude)**.** *There is no pair of sequences $a, D : \mathbb{N} \to \mathbb{N}$ with $a_n \geq 2$ for all n satisfying $D_{n+1} = a_n D_n$ and (5.1), for any $m > 0$ and any c. The same holds if the shape only begins at some index.*

Proof of the scale-primitive case. Assume first $\gcd(c, m) = 1$. Two facts about the multipliers do all the work.

Every a_j shares a prime with m. Suppose $\gcd(a_j, m) = 1$. Then m is invertible modulo a_j, so some n has $c + nm \equiv 0$, that is $a_j \mid C_n$; and $a_j \mid D_n$ for every $n > j$, since D is multiplicative with a_j among its factors. Choosing such an n beyond j and reading (5.1) modulo a_j gives $a_j \mid m$, contradicting $\gcd(a_j, m) = 1$ and $a_j \geq 2$.

A prime divisor of m occurs in at most one a_j. Suppose $p \mid m$ and $p \mid a_j$. For $n > j$ we have $p \mid D_n$, so (5.1) gives $(a_n - 1)C_n \equiv 0 \pmod{p}$. Now $C_n = c + nm \equiv c$, and $p \nmid c$ because $p \mid m$ and $\gcd(c, m) = 1$; hence $p \mid a_n - 1$, so $p \nmid a_n$. Thus p cannot divide any later multiplier.

The two facts are incompatible. Each of the infinitely many multipliers needs a prime divisor of m, and each prime divisor of m serves at most one multiplier, while m has only finitely many prime divisors. □

The scale-primitive case is the scale-primitive exclusion. The special case $m = c = 1$, where m has no prime divisors at all and the first fact is immediately contradictory, is recorded separately as the normalised exclusion.

Removing the scale. For general c, put $g = \gcd(c, m)$. Equation (5.1) shows $g \mid D_n$ for every n, so dividing D, c and m by g leaves a system of the same shape with coprime data, which the previous case excludes. □

Formalised as the exclusion at every scale, and the eventual form, obtained by shifting the orbit to the first constant index, as the eventual exclusion.

What this section proves, and what it does not. It excludes a centred state that is eventually constant and negative, at every magnitude and every scale. It says nothing about a negative state whose magnitude changes, and the argument genuinely uses constancy: the finiteness of the set of prime divisors of m is what the pigeonhole consumes, and a varying magnitude supplies a fresh set at each index.

6 Periodic negative magnitude

The next case allows the magnitude to vary, provided it repeats. Write $e_n = -E_n > 0$ for the magnitude, so $C_{n+1} = C_n + e_n$, and suppose e has period h and C gains a drift $M > 0$ over one period.

Three lemmas replace the two facts of Section 5. The first is that multipliers persist: every a_j divides every strictly later denominator state, the persistence of a multiplier. The second is a lock: a prime already dividing D and also dividing the current multiplier is forced into the next tail state, the one-step lock, and once present in both D and C it stays in every later D, C and magnitude, the persistence of a lock. The third transports a lock backwards through the period: if a divisor of the drift M occurs in two multipliers, it is a common divisor of C_0 and of every magnitude, the repeated-divisor transport, using that both the period relation and the drift iterate over any number of periods, the iterated period and the iterated drift.

Theorem 6.1 (no periodic negative magnitude). *Let $a, D, C, e : \mathbb{N} \to \mathbb{N}$ with $a_n \geq 2$, $e_n > 0$ and $e_n < a_n$ for every n, satisfying*

$$D_{n+1} = a_n D_n, \qquad C_{n+1} = C_n + e_n, \qquad D_n + e_n = (a_n - 1)C_n,$$

and suppose $e_{n+h} = e_n$ and $C_{n+h} = C_n + M$ for some $h > 0$ and $M > 0$. This is impossible.

Proof. Strong induction on M. If no prime divisor of M is a common divisor of C_0 and of every magnitude, the repeated-divisor transport says each prime divisor of M occurs in at most one multiplier; combined with the argument of Section 5, applied phase by phase, pigeonhole on the finitely many prime divisors of M gives a contradiction. Otherwise some prime p divides M, C_0 and every magnitude; the shape equation then puts p into every denominator state, so the whole orbit may be divided by p. Every hypothesis survives and the drift becomes $M/p < M$. □

The phase-primitive case is the phase-primitive exclusion, the induction is the exclusion at every scale, and the eventual form is the eventual exclusion.

The regime hypothesis. The bound $e_n < a_n$ is a genuine restriction and not a normalisation. It is the range in which a multiplier cannot divide its own phase magnitude, which is what the lock argument needs. It holds in the intended reading, where e_n is a centred representative and a_n grows quadratically, but it is assumed here and not derived. The drift hypothesis $M > 0$ is also used: a periodic magnitude with zero drift would be identically zero, which is the case already settled by Section 4.

7 A coprimality barrier

Periodicity is still a strong assumption. Removing it needs a different kind of obstruction, and the one used here is a counting argument about where an integer sequence with bounded upward steps must land.

Lemma 7.1 (shifted blocks of consecutive multiples). *Let $m_0, \ldots, m_{B-1}$ be pairwise coprime and at least 2. For every bound there is a t beyond it with $m_i \mid t + i$ for each $i < B$.*

Proof. Chinese remaindering: solve $t \equiv -i \pmod{m_i}$ simultaneously, then add multiples of $\prod_i m_i$ to pass the bound. □

Formalised as the shifted consecutive multiples.

Theorem 7.2 (bounded rise cannot avoid fresh moduli). *Let $u : \mathbb{N} \to \mathbb{N}$ tend to infinity with $u_{n+1} \leq u_n + B$ for a fixed $B > 0$. Then u cannot permanently avoid infinitely many fresh pairwise coprime moduli: there is no family of pairwise coprime $m_i \geq 2$, one for each index, such that $m_i \nmid u_t$ whenever $i < t$.*

Proof. Take B of the moduli, from indices beyond any chosen point, and use Lemma 7.1 to find t far out with the ith of them dividing $t + i$. The interval $[t, t + B)$ has length B, and u climbs by at most B per step while tending to infinity, so some u_s lands in it. Then $u_s = t + i$ for some $i < B$, and the ith modulus divides u_s, against avoidance. □

Formalised as the bounded-rise barrier. The two hypotheses pull against each other exactly: divergence forces u to travel arbitrarily far, and bounded rise forbids it from stepping over a window of width B.

The barrier applies to the problem because a *reduced* exact tail supplies the coprime moduli for free. Call (u, v) a reduced exact tail with multipliers a when $\gcd(u_n, v_n) = 1$,

$$u_{n+1} + v_n = a_n u_n, \qquad v_{n+1} = a_n v_n.$$

Proposition 7.3 (reduced tails are pairwise coprime)**.** *In a reduced exact tail, a_n is coprime to v_n; the multipliers at distinct indices are pairwise coprime; and every earlier multiplier is coprime to every later numerator.*

These are the step coprimality, the pairwise coprimality, and the whole-modulus avoidance. Feeding them to Theorem 7.2 gives the form used later: there is no reduced exact tail whose numerator tends to infinity with a uniformly bounded upward increment, the reduced bounded-rise exclusion, together with its eventual form, the eventual reduced exclusion.

Passing to a reduced tail. The orbit of Section 2 is not reduced, so one more step is needed: the common factor must stop changing. The tail gcd $\gcd(C_n, D_n)$ divides its successor, the gcd monotonicity, and at an index where the centred state is negative it is exactly $\gcd(C_n, e_n)$, the gcd at a negative index. A positive divisibility chain whose values are bounded along a cofinal set is eventually constant, the chain stabilisation; hence cofinally bounded negative magnitudes make the tail gcd stabilise, the gcd stabilisation. Beyond that point the orbit divided by the stable gcd is reduced, and Theorem 7.2 applies to it.

8 The bounded negative part

Assembling the previous section gives the strongest statement of this note. Its hypotheses are listed in full, because two of them are analytic inputs that are assumed here rather than derived from the growth condition of Problem 1.1.

Theorem 8.1 (bounded negative part)**.** *Let $a, C, D : \mathbb{N} \to \mathbb{N}$ and $E : \mathbb{N} \to \mathbb{Z}$ satisfy*

1. *$a_n > 1$ and $C_n > 0$ for every n;*
2. *the exact dynamics $C_{n+1} + D_n = a_n C_n$ and $D_{n+1} = a_n D_n$;*
3. *$E_n = \operatorname{ctr}(a_n, D_n, C_n)$ for every n;*
4. eventual strict centring*: $|E_n| < C_n$ for all large n;*
5. eventually bounded negative part*: $-B \le E_n$ for all large n, for some B;*
6. normalised vanishing*: for every K there is an N with $K\,|E_n| < C_n$ for all $n \ge N$.*

Then $E_n = 0$ for all sufficiently large n.

Proof. Suppose not. By Theorem 3.4 and its contrapositive form, E is nowhere zero beyond the centring threshold. If E is negative cofinally, then (5) bounds those magnitudes, so the tail gcd stabilises and the orbit may be taken reduced past that point.

Hypothesis (5) with Proposition 2.1 gives $C_{n+1} \le C_n + B$, a bounded rise; hypothesis (6) with E nowhere zero gives $C_n \to \infty$. A reduced exact tail with a divergent numerator and bounded rise is excluded by Theorem 7.2. If instead E is eventually positive, the descent of Theorem 4.1 applies and forces E to vanish, contradicting nowhere-vanishing. □

Formalised as the bounded-negative-part rigidity, with the eventual form, in which the centring and the bound need only hold from some index, as the eventual bounded-negative-part rigidity. The two bridges it uses are the divergence from normalised vanishing and the exclusion of cofinally bounded negative magnitudes, the latter over the tail-divergence form.

Corollary 8.2. *Under the hypotheses of Theorem 8.1, together with $C_{n+1} \neq 0$ for all large n, the multipliers satisfy $a_{n+1} = a_n^2 - a_n + 1$ for all sufficiently large n.*

Proof. Theorem 8.1 then Theorem 3.3. □

What is assumed, and what that costs. Hypotheses (1)–(3) are the state system. Hypothesis (4) says the centred state is a strict centred representative, and (6) says $|E_n|/C_n \to 0$ in a division-free form. Neither is derived here from $a_{n+1}/a_n^2 \to 1$, and no argument in the formal source connects them to Problem 1.1. Theorem 8.1 is therefore a conditional theorem about the state system, and Corollary 8.2 does not settle any case of Problem 1.1. What is unconditional in this note is the exclusion of constant and periodic negative magnitudes, Theorems 5.1 and 6.1, and the barrier itself, Theorem 7.2.

9 A finite-horizon residue reduction

One earlier tool survives, and it is worth recording that it is no longer needed for the purpose it was built for. In the normalised constant-negative template the next multiplier is forced: at index n its numerator is

$$\mathrm{num}(n, a) = (n+1)a^2 - (n+2)a + (n+3),$$

the forced numerator, and the divisor is $n+2$. The orbit survives a step when that division is exact, giving a survival predicate, the survival predicate. Deciding survival by iteration is expensive because the orbit grows doubly exponentially, and it is unnecessary.

Theorem 9.1 (factorial residue reduction)**.** *For all h and all integers $a \equiv b \pmod{(h+1)!}$, the orbit from a survives h forced updates exactly when the orbit from b does.*

Proof. Let $M(0,i) = 1$ and $M(h+1,i) = (i+2)M(h,i+1)$, an ascending factorial with $M(h,0) = (h+1)!$. The numerator is a polynomial with integer coefficients, so congruences transfer; reducing the modulus gives divisibility by $i+2$ for one exactly when for the other, and cancelling that common factor from both values and modulus leaves the inductive hypothesis at $M(h, i+1)$. □

Formalised as the factorial residue reduction, over the shrinking-modulus transport, the polynomial congruence, and the exact-division cancellation; the modulus identifications are the ascending-factorial form and the factorial value at the initial index.

The search this supported is superseded. Running it over initial states below 5000 produced forced prefixes of length 17 and no longer, which was evidence and not a proof; Theorem 5.1 now excludes the template outright, for every seed and at every scale. The reduction is retained because it is exact, and because the shrinking-modulus technique transfers to any forced orbit whose step is a polynomial division.

10 Complements and further questions

Problem #243 is open. The negative case has narrowed, and it is worth being precise about what is left.

By Theorem 3.4 a counterexample has $E_n \neq 0$ for every large n. By Theorem 4.1 it cannot have E eventually nonnegative. By Theorems 5.1 and 6.1 its negative magnitudes are neither eventually constant nor eventually periodic. By Theorem 8.1, if the analytic hypotheses (4) and (6) hold, the negative part cannot be bounded either. What remains is the following.

Problem 10.1 (unbounded negative excursions)**.** Exclude, or construct, an exact orbit whose centred state is negative infinitely often with magnitudes unbounded along that cofinal set. Every argument in Sections 5–8 consumes some finiteness: a fixed set of prime divisors, a fixed period, or a fixed bound on the negative part. An unbounded excursion supplies none of them.

Problem 10.2 (the two analytic hypotheses)**.** Derive eventual strict centring and normalised vanishing from $a_{n+1}/a_n^2 \to 1$ and rationality, or show that they do not follow. Until one or the other is done, Theorem 8.1 constrains the state system and not the original sequence.

Problem 10.3 (the two published criteria)**.** Formalise the criterion of Erdős and Straus stated in Section 1, and Duverney's conditional characterisation, under explicit analytic hypotheses. Both are cited here and neither is formalised; each is stronger than every statement in this note.

The shape of the remaining work has changed more than its difficulty. Before, the obstruction was a sign condition with no handle on it. Now the sign condition has been reduced to a growth condition on the excursions themselves, and the tool that closed the bounded case, a coprimality barrier against sequences that rise slowly and diverge, is stated in a form that does not mention the problem at all.

Statements and declarations

Artefact and data availability. The pinned formal-source revision contains the Lean sources, the fixed toolchain, and the library manifest used in the verification. This manuscript provides navigation rather than proof authority.

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A Guide to the formal sources

Each linked phrase opens its Lean declaration at the pinned source revision 4eb4c8ecfa0a. The state system, the exclusions, the barrier, and the bounded-negative-part theorem are in the first module; the residue reduction is in the second. Three distinctions are worth carrying into the source. The identification of either module with reciprocal tails is the exposition of Section 2 and is not a checked statement. The periodic exclusion assumes the regime $e_n < a_n$. And the final theorem assumes strict centring and normalised vanishing, which are hypotheses of the theorem and not consequences of the problem.

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